

TOBY CRUST

FORWARD DEPLOYED ENGINEER — CLIENT DELIVERY • AI INTEGRATION • FULL-STACK

Sydney, AU +61 425 843 487 tobycrust@gmail.com toby-crust-portfolio.vercel.app [LinkedIn /toby-crust](https://www.linkedin.com/in/toby-crust)

Software engineer who turns what **clients and non-technical teams** ask for into shipped production software. Three years delivering for government, nonprofit and enterprise-brand clients across **AI agent workflows** (Python, LLMs, Slack API), **full-stack web and integrations** (React, TypeScript, Stripe) and **on-site mobile apps** (Unity, BLE hardware), from first build through to release.

SELECTED WORK

SLIK • AI & INTERNAL TOOLING

Slack-to-VM AI agent: stakeholder requests to reviewed PRs

Built an internal pipeline that lets non-technical team members request project changes in plain language in Slack. Each request is dispatched to an AI agent that provisions a virtual machine, applies the code change, runs the test suite and alerts me for final PR review and merge. It drastically cut routine work and raised the whole team's shipping velocity.

Python

AI Agents

Slack API

VM Automation

GitHub Actions

Automated Testing

CI/CD

SLIK • NORTHERN TERRITORY GOVERNMENT

On-site AR heritage app, deployed in the field

Lead developer on a historical WWII AR app for the NT Government on iOS and Android. Content triggers on site through BLE beacons and geolocation. Unity Addressables keep the store download small and let new content ship without a full app release. The first site's success means further locations and experiences are in production now.

Unity

C#

AR Foundation

ARKit / ARCore

BLE beacons

Addressables

Localization

Convai

SLIK • FIGHTMND

National campaign donation platform

Full-stack build of the donation site for a national fundraising campaign: Stripe payment flow, supporter data capture, and a front end that held up under broadcast-driven traffic spikes.

React

TypeScript

Stripe

Netlify

\$165,575

RAISED THROUGH THE PLATFORM

8.9K

PEAK CONCURRENT VIEWERS

EXPERIENCE

Dec 2025 – present

Slik — Software Engineer, Sydney

- Deliver client projects end to end for government and nonprofit clients, from first build through App Store Connect and Play Console release.
- Built Slack-to-VM AI agent workflows that turn stakeholder requests into tested pull requests.
- Integrate client systems: Stripe checkout, mailing-list and ESP integrations, GitHub Actions data pipelines, Netlify deployments and DNS.

Oct 2024 – Oct 2025

Fabra — Software Engineer, Sydney

- Took over a departing senior engineer's workstream and shipped share-design permissions and email validation end to end, front and back.
- Wrote internal tooling that cut manual steps out of the team's day-to-day backend operations.
- Built features across Fabra's core SaaS product in React and TypeScript.

Feb – Oct 2024

Bardon Design — Junior Software Engineer, Bangkok

- Worked with the Louis Vuitton team on a gamified e-commerce experience built to lift engagement through interactive shopping.
- Built an interactive AR furniture selection app.

Summers, 2021 – 2023

Software internships — Auckland

- Built an AR Rube Goldberg machine level for GeoArGames, and a 2D arcade game embedded in the Yuri&Niels site.

SIDE PROJECTS

TickyTick assistant — LLM with a human in the loop

Gemini function-calling assistant in my local-first task manager. The model only proposes changes; the app validates them and shows them for review before anything is written.

Gemini Function calling Supabase

Hexapod robot — hardware integration

Six-legged walking robot: Arduino firmware with inverse kinematics, driven over WiFi from a three.js leg editor through a Python bridge.

C++ Python three.js ESP8266

TOOLKIT

LANGUAGES

TypeScript / JavaScript
Python
C#
C++

AI & AGENTS

AI Agents / LLMs
Function Calling
Prompt Engineering
Slack API & Webhooks
VM Automation

INTEGRATIONS

REST APIs
Stripe
ESP Integrations
Supabase
BLE Beacons

SHIP & OPERATE

GitHub Actions / CI
Automated Testing
Netlify / Vercel / DNS
App Store Connect
Google Play Console

EDUCATION

Bachelor of Design Innovation — Victoria University, 2021–2023

REFERENCES

Available on request — previous engineering leads at Fabra and Bardon Design are happy to speak.

OUTSIDE WORK

Toastmasters · Robotics · 3D printing · Snowboarding · Waterpolo